HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

SCHOOL OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

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FINAL EXAM – DATABASE PRACTICE

Duration: 75 minutes. No materials are allowed.

Full name: .... Student ID: ....

A. Database Creation: Bookstore_StudentID – 1.5 points

Create a database named Bookstore_StudentID with the following tables.

- Create all table structures: 0.5 point
- Create relationship diagram: 0.5 point
- Insert the given sample data into all tables (each table only needs to contain 1 sample record): 0.5 point

Database Schema: Primary-key attributes are <u>underlined</u>; Foreign-key attributes are italicized.

BOOK(<u>BookID</u>, Title, Author, Category, Price, Publisher)

READER(<u>ReaderID</u>, FullName, Address, Phone, DateOfBirth, ReaderType)

INVOICE(<u>InvoiceID</u>, InvoiceDate, ReaderID)

INVOICE_DETAIL(<u>InvoiceID, BookID</u>, Quantity)

Sample Data

Table: BOOK

BookID	Title	Author	Category	Price	Publisher
B001	Basic Python Programming	John Smith	IT	150000	Young Publishing House
B002	Advanced Mathematics Vol. I	Mary Johnson	Mathematics	85000	Education Press
B003	English Literature	Robert Brown	Literature	60000	Culture Press
B004	Advanced Java Programming	John Smith	IT	200000	Young Publishing House
B005	General Physics	Linda Davis	Science	95000	Education Press
B006	C++ Programming Techniques	Michael Wilson	IT	175000	National University Press
B007	World History	Emily Taylor	History	70000	Culture Press
B008	Introduction to Artificial Intelligence	John Smith	IT	220000	Young Publishing House

Table: READER

ReaderID	FullName	Address	Phone	DateOfBirth	ReaderType
R001	David Miller	12 Green Street, Hanoi	0912345678	2000-03-05	Student
R002	Emma Johnson	34 Central Avenue, Hanoi	0923456789	1999-07-12	Student
R003	Daniel Brown	56 Riverside Road, Hanoi	0934567890	1990-11-20	Lecturer
R004	Sophia Wilson	78 Hill Street, Hanoi	0945678901	2002-09-08	Student
R005	James Anderson	90 Lake View Road, Hanoi	0956789012	1985-04-15	Lecturer
R006	Olivia Taylor	22 Garden Street, Hanoi	0967890123	2001-06-30	Student

Table: INVOICE

InvoiceID	InvoiceDate	ReaderID
I001	2022-03-10	R001
I002	2022-03-15	R002
I003	2022-03-22	R001
I004	2022-05-05	R003
I005	2022-05-10	R004
I006	2022-06-20	R002
I007	2022-07-15	R001
I008	2023-09-20	R005
I009	2023-10-10	R006
I010	2023-11-25	R003

Table: INVOICE_DETAIL

InvoiceID	BookID	Quantity
I001	B001	2
I001	B004	1
I002	B002	3
I002	B005	2
I003	B001	1
I003	B006	1
I004	B003	2
I005	B004	2
I006	B002	1
I006	B007	3
I007	B001	1
I007	B008	1
I008	B005	2
I009	B001	3
I010	B003	1
I010	B008	2

B. Query Requirements – 8.5 points: Write SQL statements to answer the following requirements.

1. Display the information of readers (ReaderID, FullName, Address, Phone) who bought at least one book in the "IT" category in 2022. (1 point)

2. Display the list of books (BookID, Title, Price, Category) that had the highest total quantity sold in 2022. (1 point)

3. Display the list of readers (ReaderID, FullName, Address, Phone, DateOfBirth) who bought at least 2 books "Basic Python Programming" in the bookstore. (1 point)

4. Display the information of readers (ReaderID, FullName, Address, Phone) who bought books from at least two different categories in 2022. (1 point)

5. Display the information of books (BookID, Title, Category, Price) that were sold in both 2022 and 2023. (1 point)

6. Display VIP readers (ReaderID, FullName, Address, [Total number of purchases], [Total invoice amount]) in 2022. A VIP reader is a reader whose total invoice amount is greater than 1000000 VND and whose total number of purchases is at least 3. (1 point)

7. Create a constraint on table READER to ensure that phone numbers in column Phone must contain exactly 10 digits and must start with the digit '0'. (1 point)

8. For each year, display the category or categories with the highest total revenue. The result should include (Year, Category, TotalRevenue). (1 point)

9. For each reader, find the pair of consecutive purchases with the largest time gap. The result should include (ReaderID, FullName, PreviousInvoiceID, PreviousInvoiceDate, NextInvoiceID, NextInvoiceDate, GapInDays). (0.25 point)

10. Find the categories whose revenue in 2023 increased more than the average revenue growth of all categories that were sold in both 2022 and 2023. The result should include (Category, Revenue2022, Revenue2023, GrowthAmount, GrowthRate), where: GrowthAmount = Revenue2023 Revenue2022; GrowthRate = (Revenue2023 Revenue2022) / Revenue2022.
(0.25 point)